Task 1:

```
bodike@bodike-chaithali:~/Home$ ls
dir2  Lab3  student_id_new.txt  students_marks
bodike@bodike-chaithali:~/Home$ mkdir Assignment_01_B24CS013
bodike@bodike-chaithali:~/Home$ ls
Assignment_01_B24CS013  Lab3               students_marks
dir2                    student_id_new.txt
bodike@bodike-chaithali:~/Home$ cd Assignment_01_B24CS013/
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ ls
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat > weather_details.txt << EOF
> New York,28,65,Partly Cloudy,14
> Tokyo,31,72,Sunny,10
> London,22,80,Rainy,18
> Mumbai,30,85,Humid,12
> Paris,26,70,Clear,8
> Sydney,18,60,Windy,22
> Dubai,41,40,Hot,20
> Moscow,19,75,Over,cast,15
> Cape Town,24,55,Sunny,13
> Sao Paulo,27,68,Cloudy,11
> EOF
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ ls
weather_details.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat weather_details.txt
New York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
London,22,80,Rainy,18
Mumbai,30,85,Humid,12
Paris,26,70,Clear,8
Sydney,18,60,Windy,22
Dubai,41,40,Hot,20
Moscow,19,75,Over,cast,15
Cape Town,24,55,Sunny,13
Sao Paulo,27,68,Cloudy,11
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Q1)

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ awk -F',' ' $3>=70 && $5>=15 {print $1","$2","$4 > "storm_probability.txt"}' weather_de
tails.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat storm_probability.txt
London,22,Rainy
Moscow,19,Over
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat weather_details.txt
New York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
London,22,80,Rainy,18
Mumbai,30,85,Humid,12
Paris,26,70,Clear,8
Sydney,18,60,Windy,22
Dubai,41,40,Hot,20
Moscow,19,75,Over,cast,15
Cape Town,24,55,Sunny,13
Sao Paulo,27,68,Cloudy,11
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Q2)

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ sort -t',' -k2 -n storm_probability.txt| head -n 1 | cut -d',' -f1
Moscow
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ grep -v "^$(sort -t',' -k2 -n storm_probability.txt | head -n 1 | cut -d',' -f1)," weat
her_details.txt >> temp && mv temp weather_details.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat weather_details.txt
New York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
London,22,80,Rainy,18
Mumbai,30,85,Humid,12
Paris,26,70,Clear,8
Sydney,18,60,Windy,22
Dubai,41,40,Hot,20
Cape Town,24,55,Sunny,13
Sao Paulo,27,68,Cloudy,11
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Q3)

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ awk -F',' '{if($2<=25) { wind =($5>=15) ? "HIGH WIND" : "LOW WIND"; print $1":"$2":"win
d;} }' weather_details.txt | sort -t':' -k2 -nr >> cold_weather_high_wind.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat cold_weather_high_wind.txt
Cape Town:24:LOW WIND
London:22:HIGH WIND
Sydney:18:HIGH WIND
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Q4)

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ awk -F',' '{if($2<=25) { wind =($5>=15) ? "HIGH WIND" : "LOW WIND"; print $1":"$2":"win
d;} }' weather_details.txt | sort -t':' -k2 -nr >> cold_weather_high_wind.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat cold_weather_high_wind.txt
Cape Town:24:LOW WIND
London:22:HIGH WIND
Sydney:18:HIGH WIND
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat weather_details.txt                                                          Ne
w York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
London,22,80,Rainy,18
Mumbai,30,85,Humid,12
Paris,26,70,Clear,8
Sydney,18,60,Windy,22
Dubai,41,40,Hot,20
Cape Town,24,55,Sunny,13
Sao Paulo,27,68,Cloudy,11
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ sed -i "/^$(awk -F':' '$3=="HIGH WIND" {print $1}' cold_weather_high_wind.txt | tail -n
 1)/d" weather_details.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat weather_details.txt
New York,28,65,Partly Cloudy,14
Tokyo,31,72,Sunny,10
London,22,80,Rainy,18
Mumbai,30,85,Humid,12
Paris,26,70,Clear,8
Dubai,41,40,Hot,20
Cape Town,24,55,Sunny,13
Sao Paulo,27,68,Cloudy,11
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Task 2:

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ for i in {1..10}; do echo "This is line$i."; done > file1.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat file1.txt
This is line1.
This is line2.
This is line3.
This is line4.
This is line5.
This is line6.
This is line7.
This is line8.
This is line9.
This is line10.
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

```
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ ls
cold_weather_high_wind.txt  file1.txt  storm_probability.txt  weather_details.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ vi tail.c
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ ls
cold_weather_high_wind.txt  file1.txt  storm_probability.txt  tail.c  weather_details.txt
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ gcc tail.c -o tail
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ ./tail 3 file1.txt
This is line8.
This is line9.
This is line10.
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$ cat tail.c
#include <stdio.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s number_of_lines filename\n", argv[0]);
        return 1;
    }
    int n = 0;
    int i = 0;
    while (argv[1][i] != '\0') {
            if (argv[1][i] < '0' || argv[1][i] > '9') {
                    printf("Invalid number: %s\n", argv[1]);
                    return 1;
    }
    n = n * 10 + (argv[1][i] - '0');
    i++;
}

    if (n <= 0 || n > 1000) {
        printf("Please enter a number between 1 and 1000\n");
        return 1;
    }
    FILE *fp = fopen(argv[2], "r");
    if (fp == NULL) {
        printf("Error in opening the file %s.\n", argv[2]);
        return 1;
    }
```

```
                    return 1;
    }
    n = n * 10 + (argv[1][i] - '0');
    i++;
}

    if (n <= 0 || n > 1000) {
        printf("Please enter a number between 1 and 1000\n");
        return 1;
    }
    FILE *fp = fopen(argv[2], "r");
    if (fp == NULL) {
        printf("Error in opening the file %s.\n", argv[2]);
        return 1;
    }
    char lines[1000][256];
    int count = 0;

    while (fgets(lines[count], 256, fp) != NULL) {
        count++;
        if (count == 1000) {
            break;
        }
    }
    fclose(fp);

    int start = 0;
    if (count > n) {
        start = count - n;
    }

    for (int i = start; i < count; i++) {
        printf("%s", lines[i]);
    }

    return 0;
}
bodike@bodike-chaithali:~/Home/Assignment_01_B24CS013$
```

Task 3:

Q1)

```
bodike@bodike-chaithali:~/Home$ ls
Assignment_01_B24CS013  Lab3                students_marks
dir2                    student_id_new.txt
bodike@bodike-chaithali:~/Home$ mkdir EmployeeRecords
bodike@bodike-chaithali:~/Home$ ls
Assignment_01_B24CS013  EmployeeRecords  student_id_new.txt
dir2                    Lab3             students_marks
bodike@bodike-chaithali:~/Home$ cd EmployeeRecords/
bodike@bodike-chaithali:~/Home/EmployeeRecords$ ls
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q2)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ ls
bodike@bodike-chaithali:~/Home/EmployeeRecords$ touch emp_basic.txt emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ ls
emp_basic.txt  emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q3)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat > emp_basic.txt << EOF
> 101,John,Manager
> 102,Alice,Developer
> 103,Bob,Analyst
> EOF
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Manager
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q4)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat > emp_salary.txt << EOF
> 101,85000
> 102,70000
> 103,65000
> EOF
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt

101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q5)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ ls -l emp_basic.txt emp_salary.txt
-rw-rw-r-- 1 bodike bodike 53 Aug 10 22:25 emp_basic.txt
-rw-rw-r-- 1 bodike bodike 31 Aug 10 22:26 emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ chmod 600 emp_basic.txt emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ ls -l emp_basic.txt emp_salary.txt
-rw------- 1 bodike bodike 53 Aug 10 22:25 emp_basic.txt
-rw------- 1 bodike bodike 31 Aug 10 22:26 emp_salary.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q6)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Manager
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cut -d, -f2 emp_basic.txt
John
Alice
Bob
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Manager
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q7)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ sort -t, -k2 -nr emp_salary.txt
101,85000
102,70000
103,65000

bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt

101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q8)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt

101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Manager
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ join -t, -1 1 -2 1 emp_basic.txt emp_salary.txt >
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat merged.txt
101,John,Manager,85000
102,Alice,Developer,70000
103,Bob,Analyst,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q9)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ join -t, -1 1 -2 1 emp_basic.txt emp_salary.txt > merged.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat merged.txt
101,John,Manager,85000
102,Alice,Developer,70000
103,Bob,Analyst,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cut -d, -f 2,4 merged.txt
John,85000
Alice,70000
Bob,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat merged.txt
101,John,Manager,85000
102,Alice,Developer,70000
103,Bob,Analyst,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q10)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Manager
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ sed -i 's/Manager/Team_lead/g' emp_basic.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q11)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ wc merged.txt
 3  3 71 merged.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q12)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat > emp_salary.txt << EOF
101,85000
102,70000
103,65000
EOF
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ paste -d: emp_basic.txt  emp_salary.txt
101,John,Team_lead:101,85000
102,Alice,Developer:102,70000
103,Bob,Analyst:103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q13)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ grep er$ emp_basic.txt
102,Alice,Developer
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q14)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$  join -t, -1 1 -2 1 emp_basic.txt emp_salary.txt | awk -F',' '$4 > 65000' | sort -t, -k4 -nr >
 high_earners.txt
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat high_earners.txt
101,John,Team_lead,85000
102,Alice,Developer,70000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Q15)

```
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cut -d, -f2 emp_basic.txt | grep -E '^[A-Z][a-zA-Z]{3,}$'
John
Alice
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_basic.txt
101,John,Team_lead
102,Alice,Developer
103,Bob,Analyst
bodike@bodike-chaithali:~/Home/EmployeeRecords$ cat emp_salary.txt
101,85000
102,70000
103,65000
bodike@bodike-chaithali:~/Home/EmployeeRecords$
```

Task 4:

Q1)

```
bodike@bodike-chaithali:~/Home$ ls
Assignment_01_B24CS013  EmployeeRecords  student_id_new.txt
dir2                    Lab3             students_marks
bodike@bodike-chaithali:~/Home$ mkdir UniversityDB
bodike@bodike-chaithali:~/Home$ ls
Assignment_01_B24CS013  EmployeeRecords  student_id_new.txt  UniversityDB
dir2                    Lab3             students_marks
bodike@bodike-chaithali:~/Home$ cd UniversityDB/
bodike@bodike-chaithali:~/Home/UniversityDB$ cat > student_academic.txt << EOF
> 101,Rahul,CS,78,85,90
> 102,Aditi,IT,55,60,58
> 103,Neha,EE,88,92,80
> 104,Arjun,CS,35,40,45
> 105,Meera,IT,95,96,94
> EOF
bodike@bodike-chaithali:~/Home/UniversityDB$ ls
student_academic.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_academic.txt
101,Rahul,CS,78,85,90
102,Aditi,IT,55,60,58
103,Neha,EE,88,92,80
104,Arjun,CS,35,40,45
bodike@bodike-chaithali:~/Home/UniversityDB$ cat > student_stipend.txt << EOF
> 101,1500
> 102,0
> 103,2000
> 104,0
> 105,2500
> EOF
bodike@bodike-chaithali:~/Home/UniversityDB$ ls
student_academic.txt  student_stipend.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_stipend.txt
101,1500
102,0
103,2000
104,0
105,2500
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q2) and Q3)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ ls -l student_stipend.txt student_academic.txt
-rw-rw-r-- 1 bodike bodike 109 Aug 10 23:06 student_academic.txt
-rw-rw-r-- 1 bodike bodike  39 Aug 10 23:07 student_stipend.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ chmod 640 student_stipend.txt student_academic.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ ls -l student_stipend.txt student_academic.txt
-rw-r----- 1 bodike bodike 109 Aug 10 23:06 student_academic.txt
-rw-r----- 1 bodike bodike  39 Aug 10 23:07 student_stipend.txt
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q4)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ date +"%Y-%m-%d %H:%M:%S"
2025-08-10 23:12:07
```

Q5)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ ls
student_academic.txt  student_stipend.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_academic.txt
101,Rahul,CS,78,85,90
102,Aditi,IT,55,60,58
103,Neha,EE,88,92,80
104,Arjun,CS,35,40,45
105,Meera,IT,95,96,94
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_stipend.txt
101,1500
102,0
103,2000
104,0
105,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ join -t, -1 1 -2 1 student_academic.txt student_stipend.txt > merged.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,CS,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,CS,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ ls
merged.txt  student_academic.txt  student_stipend.txt
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q6)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ awk -F',' '{total=$4+$5+$6; print $2","total}' merged.txt
Rahul,253
Aditi,173
Neha,260
Arjun,120
Meera,285
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q7)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ awk -F',' '{total=$4+$5+$6; print total","$0}' merged.txt | sort -t',' -k2 -nr | head -n 1 | cut
-d',' -f2-
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,CS,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,CS,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q8)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_academic.txt
101,Rahul,CS,78,85,90
102,Aditi,IT,55,60,58
103,Neha,EE,88,92,80
104,Arjun,CS,35,40,45
105,Meera,IT,95,96,94
bodike@bodike-chaithali:~/Home/UniversityDB$ awk -F',' '$4 < 40 || $5 < 40 || $6 < 40 {print $2","$3}' student_academic.txt
Arjun,CS
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q9)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,CS,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,CS,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ sed -i 's/CS/Computer_Science/g' merged.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,Computer_Science,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,Computer_Science,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q10)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,Computer_Science,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,Computer_Science,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ awk -F',' '$7 > 2000 {print $2}' merged.txt
Meera
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q11)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ cat merged.txt
101,Rahul,Computer_Science,78,85,90,1500
102,Aditi,IT,55,60,58,0
103,Neha,EE,88,92,80,2000
104,Arjun,Computer_Science,35,40,45,0
105,Meera,IT,95,96,94,2500
bodike@bodike-chaithali:~/Home/UniversityDB$ join -t, -1 1 -2 1 student_academic.txt student_stipend.txt | awk -F',' '$7>0' | sort -t, -k7 -nr
 | cut -d, -f 2,7
Meera,2500
Neha,2000
Rahul,1500
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q12)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ join -t, -1 1 -2 1 student_academic.txt student_stipend.txt | awk -F',' '$7>0' | sort -t, -k7 -nr
 | cut -d, -f 2,7
Meera,2500
Neha,2000
Rahul,1500
bodike@bodike-chaithali:~/Home/UniversityDB$ join -t, -1 1 -2 1 student_academic.txt student_stipend.txt | awk -F',' '$7>0' | sort -t, -k7 -nr
 | cut -d, -f 2,7 > stipend_list.txt
bodike@bodike-chaithali:~/Home/UniversityDB$ cat stipend_list.txt
Meera,2500
Neha,2000
Rahul,1500
bodike@bodike-chaithali:~/Home/UniversityDB$
```

Q13)

```
bodike@bodike-chaithali:~/Home/UniversityDB$ cut -d, -f2 student_academic.txt |  grep -E '^N[a-zA-Z]{3,}$'
Neha
bodike@bodike-chaithali:~/Home/UniversityDB$ cat student_academic.txt
101,Rahul,CS,78,85,90
102,Aditi,IT,55,60,58
103,Neha,EE,88,92,80
104,Arjun,CS,35,40,45
105,Meera,IT,95,96,94
bodike@bodike-chaithali:~/Home/UniversityDB$ \
```